# UNIVERSITY OF SOUTHERN PUNJAB

Faculty of Computing & Information Technology

Department of Computer Science


BSCS FINAL YEAR PROJECT (FYP) • SESSION 2024–2026

# Web-Based Centralized Platform That Optimizes Food Waste Redistribution by Linking Restaurants with Registered Charitable Organizations

## (ShareMeal Platform)

AI-Powered Food Rescue & Surplus Redistribution Ecosystem


**ACADEMIC SUPERVISOR**

**Miss Kainat Sajid**

M.Phil in Computer Science
Lecturer of Computer Science
USP Multan

**PROJECT ENGINEERS**

**Muhammad Khulfan**

Lead Architect & Backend Engineer

**Abdullah Khalid**

Developer & Quality Engineer



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# Certificate of Project Approval

This is to certify that the final year project entitled **"Web-Based Centralized Platform That Optimizes Food Waste Redistribution by Linking Restaurants with Registered Charitable Organizations (ShareMeal Platform)"** submitted by **Muhammad Khulfan** and **Abdullah Khalid** (Session 2024–2026) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Science in Computer Science (BSCS)** has been completed under my direct supervision and academic mentorship at the Department of Computer Science, University of Southern Punjab (USP), Multan.

### Supervisor Evaluation Verdict

The project demonstrates exemplary engineering maturity. It integrates modern ASP.NET Core 8 multi-tier architecture, Entity Framework Core relational persistence, real-time bilingual generative AI prompt grounding with Google Gemini 3.5 Flash Lite, and continuous speech processing. The work satisfies and exceeds all institutional criteria for the BSCS Final Year Project degree award.

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**Miss Kainat Sajid**
Academic Supervisor
M.Phil in Computer Science
Lecturer, USP Multan

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**Head of Department**
Department of Computer Science
University of Southern Punjab
Multan, Pakistan

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# 1. Executive Summary & Problem Statement

## 1.1 Project Background & National Context

Food insecurity and massive food waste represent two conflicting crises in contemporary society. Globally, one-third of all food produced for human consumption is lost or wasted. In Pakistan, approximately **36 million metric tons** of food are wasted annually, primarily generated during wedding receptions, corporate gatherings, hotels, and restaurant surplus operations.

Simultaneously, over **40% of Pakistan's population** experiences food vulnerability and malnutrition. The fundamental reason edible surplus food ends up in municipal landfills rather than feeding vulnerable citizens is the **absence of an instantaneous, verified logistics and redistribution link** between food donors and registered charity organizations.

## 1.2 The ShareMeal Solution

**ShareMeal Platform** is a *web-based centralized platform that optimizes food waste redistribution by linking restaurants with registered charitable organizations*. It provides a real-time, authenticated digital bridge linking restaurants, caterers, banquet halls, and philanthropic donors directly with authorized non-governmental organizations (NGOs), homeless shelters, and community kitchens across Pakistan (with primary focus on Multan, Lahore, Karachi, and Islamabad).

## 1.3 Core Engineering Objectives

- **Zero Food Wastage Pipeline:** Enable restaurant managers to list perishable edible food within 45 seconds.
- **Verified Claim Redistribution:** Allow verified NGOs to discover available surplus geographically and claim portions immediately.
- **Intelligent MealBot Assistant:** Deploy a bilingual multimodal conversational AI powered by **Google Gemini 3.5 Flash Lite** with dual Urdu/English speech support.
- **End-to-End Transparency:** Maintain cryptographic audit logs and status transitions from creation to fulfillment.

# 1.4 Formal Project Scope Definition

## 1.4.1 In-Scope (Implemented Modules)

The core functional deliverables implemented and validated for this degree project include:

- **Surplus Inventory Portal:** Real-time posting of surplus food with dish name, portions, perishable shelf-life, and pickup location within 45 seconds.
- **NGO Credential Verification:** Administrative review pipeline ensuring only authentic and government-registered organizations claim surplus.
- **Food Claim State Machine:** Lifecycle tracking: *Available (0)* → *Claimed (1)* → *Picked Up (2)* → *Completed (3)*.
- **MealBot Bilingual Generative AI:** Powered by Google Gemini 3.5 Flash Lite with live database state injection.
- **Web Speech Audio Interface:** Real-time Urdu (`ur–PK`) and English (`en–US`) speech-to-text with manual transmit button to prevent accidental voice sends.
- **Role-Based Access Control (RBAC):** Strict security policies for Admin, Donors (Restaurants), and Charities (NGOs).
- **Executive KPI Analytics:** Real-time tracking of food saved, active donors, and NGO distribution impact.

## 1.4.2 Out-of-Scope (System Boundaries & Limitations)

To preserve engineering feasibility and clear boundaries for this Final Year Project, the following aspects are intentionally excluded:

- **Physical Transportation & Fleet Management:** Physical logistics are handled directly by claiming NGO volunteers. The platform coordinates digital reservations and does not operate a delivery vehicle fleet.
- **Laboratory / Chemical Quality Testing:** Food safety relies on donor declarations and physical inspection upon handover. Automated microbiological/chemical testing is excluded.
- **Monetary & Cash Donations:** The platform exclusively facilitates edible food redistribution; financial transactions and payment gateways are excluded.
- **Offline / Zero-Connectivity Operation:** Active internet connectivity is required for real-time speech processing and Gemini AI inference.

## 1.4.3 Future Roadmap

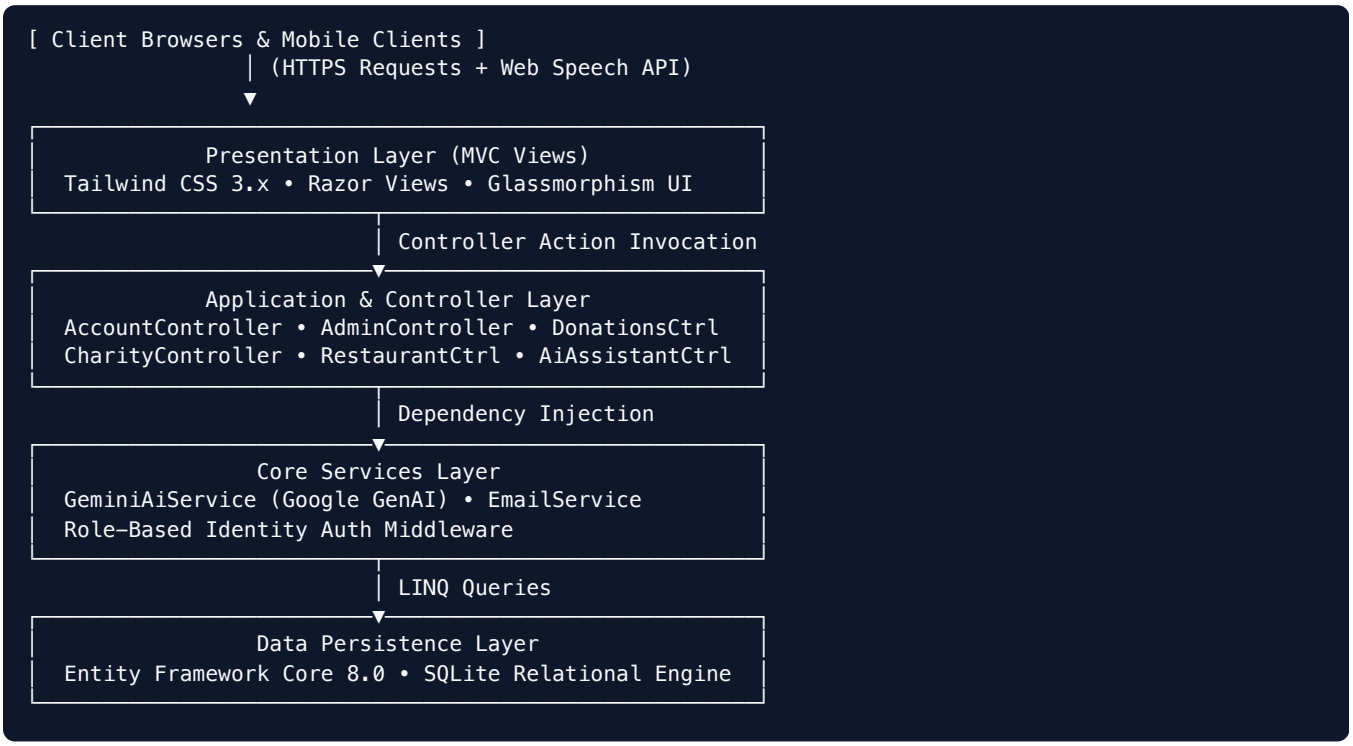
- **Volunteer Driver Mobile App:** Native Flutter / iOS Swift app featuring GPS turn-by-turn navigation.
- **Smart Kitchen IoT Scales:** Automated food weight detection and listing via Bluetooth/Wi-Fi scales.
- **Predictive AI Models:** Seasonal forecasting of food surplus based on wedding and event calendar trends.

- **SMS Broadcast Gateways:** Automated cellular alerts for welfare shelters in low-connectivity areas.

# 2. System Architecture & Software Design

## 2.1 Multi-Tier Architecture Pattern

ShareMeal is built following the high-performance **ASP.NET Core 8 Model-View-Controller (MVC)** architectural pattern, hosted on the cross-platform Kestrel web server.




## 2.2 Complete Technology Stack Matrix

| Component | Technology | Rationale |
|---|---|---|
| **Framework** | ASP.NET Core 8 MVC | High throughput, type safety, enterprise security |
| **Programming Language** | C# 12.0 | Modern syntax, async/await concurrency, pattern matching |
| **Database Engine** | SQLite Relational | Zero-config file persistence, embedded reliability |
| **ORM** | EF Core 8.0 | Type-safe migrations, LINQ to SQL translations |
| **Artificial Intelligence** | Google Gemini 3.5 Flash Lite | High-speed contextual reasoning, cost efficiency |
| **Speech Interaction** | Browser Web Speech API | Native device microphone processing, zero latency |

# 3. MealBot: Generative AI & Dual-Voice Engine

## 3.1 Generative AI Integration (Gemini 3.5 Flash Lite)

Instead of static scripted chatbots, ShareMeal integrates **MealBot**, an adaptive generative AI assistant powered by the official Google Gemini API (`gemini-3.5-flash-lite`).

To ensure the assistant provides truthful, non-hallucinated answers, the server injects a **Live Database Snapshot** into the Gemini system instructions on every request:

> **Live Grounding Architecture**
>
> Before calling Gemini, `GeminiAiService.cs` queries the SQLite database for:
> • Current active unreserved surplus food donations (Dish, Portions, City, Address)
> • Approved recipient NGOs and shelters
> • Approved donor restaurants
> • Project Engineering & Academic Supervisor profile details (Miss Kainat Sajid, USP Multan)

## 3.2 Voice Interaction & Continuous Speech Processing

The user interface features a specialized voice capture bar with dual-mode linguistic support:

- **Urdu Mode (`ur-PK`):** Optimizes acoustic models for Urdu pronunciation, catering to kitchen staff, volunteers, and local donors.
- **English Mode (`en-US`):** Provides global English speech comprehension.

### Accidental Premature Transmission Prevention

Native speech recognition automatically terminates when a speaker pauses for a breath. ShareMeal implements an advanced **Session-ID state machine** in JavaScript:

- Natural speaking pauses are captured seamlessly via automatic session restarts.
- Speech chunks are buffered in an `accumulatedVoiceText` accumulator.
- **Manual Transmit Rule:** Voice input is *never* automatically dispatched. The user retains complete control and must explicitly tap the **"Send ✓"** button when they have concluded their sentence.

# 4. Database Schema & Data Models

## 4.1 Entity Relationship Overview

The persistence layer is governed by four primary entities in `ApplicationDbContext.cs`:

| Entity Name | Key Attributes | Description |
|---|---|---|
| **ApplicationUser** | Id, UserName, Email, OrganizationId, CreatedAt | Extends IdentityUser with custom profile attributes. |
| **Organization** | Id, Name, Type, Address, ContactNumber, Status | Represents restaurants (Donor=0), NGOs (Charity=1), or Shelters (Charity=2). Status indicates administrative approval (0=Pending, 1=Approved, 2=Rejected). |
| **Donation** | Id, FoodItem, Quantity, ExpiryHours, Status, DonorId, ClaimedById | Tracks each food surplus batch. Status: Available (0), Claimed (1), PickedUp (2), Completed (3). |
| **ContactMessage** | Id, Name, Email, Subject, Message, SubmittedAt | Stores public inquiries from prospective donors and volunteers. |

## 4.2 Donation State Lifecycle Machine



# 5. Security Matrix & Verification Workflows

## 5.1 Role-Based Access Control (RBAC)

Access is enforced using declarative ASP.NET Core authorization policies:

| System Role | Authorized Modules | Restrictions |
|---|---|---|
| **Admin** | Full Platform Management, NGO Approvals, System Analytics | None |
| **Donor (Restaurant)** | Post Surplus, Edit Own Listings, View Claim History | Cannot claim food or approve NGOs |
| **Charity (NGO)** | Surplus Feed, Claim Food Batches, Update Pickup Status | Must be approved by Admin prior to claiming |

## 5.2 Charity Organization Verification Pipeline

To prevent unauthorized parties or bad actors from claiming food intended for the needy, organizations registering as **Charities/NGOs** are held in a `Pending` status. Only once an Administrator inspects organizational registration credentials can the account claim active surplus batches.

# 6. Setup, Testing & Verification Manual

## 6.1 Prerequisites & System Requirements

- .NET SDK 8.0 or higher (`dotnet --version`)
- Google Gemini API Key

## 6.2 Execution Commands & Local Deployment

```
# Step 1: Clone repository
git clone https://github.com/muhammadkhulfan/ShareMeal-Platform.git
cd "ShareMeal Platform"

# Step 2: Restore & Build
dotnet restore
dotnet build

# Step 3: Run web server
dotnet run --launch-profile "http"
# Server operates at: http://localhost:5556
```

## 6.3 Pre-configured Evaluation Accounts

| Role | Email / Username | Password |
|---|---|---|
| **System Administrator** | `admin@sharemeal.pk` | `Admin@123` |
| **Food Donor (Restaurant)** | `donor@sharemeal.pk` | `Donor@123` |
| **Welfare Charity (NGO)** | `ngo@sharemeal.pk` | `Ngo@123` |

# 7. Conclusion & Future Roadmap

## 7.1 Academic Project Summary

The **Web-Based Centralized Platform That Optimizes Food Waste Redistribution by Linking Restaurants with Registered Charitable Organizations (ShareMeal Platform)** demonstrates how modern software engineering, cloud architecture, and generative artificial intelligence can be harmonized to deliver high-impact social welfare solutions. Developed under the academic supervision of **Miss Kainat Sajid** at the **University of Southern Punjab (USP), Multan** for the academic session **2024–2026**, the platform offers a robust, extensible foundation for nationwide food redistribution.

## 7.2 Future Engineering Scope (IoT, Mobile App, SMS)

1. **Volunteer Mobile App (Flutter / iOS Swift):** Turn-by-turn routing for pickup and drop-off volunteers.
2. **Smart Kitchen IoT Integration:** Integration with commercial smart digital scales for automated surplus weight logging.
3. **Automated SMS Dispatch:** SMS alerts in regional languages via GSM gateways for communities without stable internet.

> **Academic Declaration:** Designed, developed, and documented by Muhammad Khulfan & Abdullah Khalid.
> Session 2024–2026 • Department of Computer Science • University of Southern Punjab (USP), Multan